

Data Handling and Visualization in R

Assignment Solutions

Geospatial, Network, Temporal, Multivariate and Integrated Data Visualization

1. Geospatial Data Visualization

City	Latitude	Longitude	Population (Millions)
Chennai	13.08	80.27	11
Bangalore	12.97	77.59	13
Hyderabad	17.38	78.48	10
Mumbai	19.07	72.87	20
Delhi	28.61	77.21	32
Pune	18.52	73.85	7
Kolkata	22.57	88.36	15
Ahmedabad	23.02	72.57	8
Jaipur	26.91	75.79	4
Kochi	9.93	76.26	2

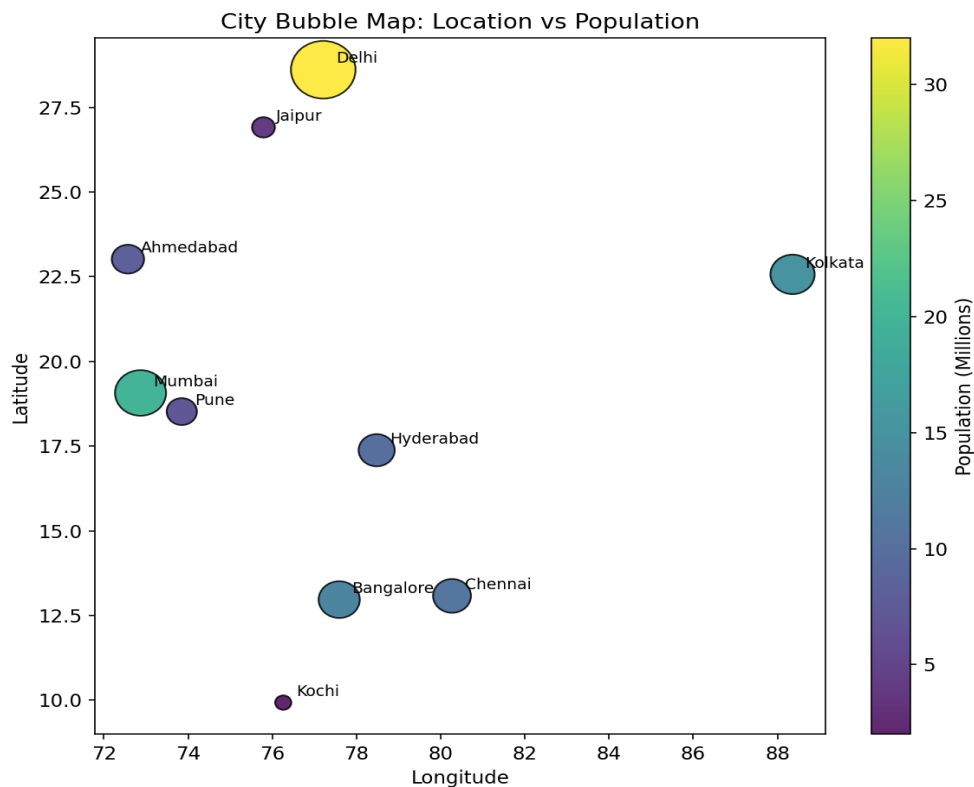
R code (uses ggplot2) for Q1-Q4 is combined below since all four build on the same plot object.

```
library(ggplot2)

cities <- data.frame(
  City = c('Chennai','Bangalore','Hyderabad','Mumbai','Delhi',
           'Pune','Kolkata','Ahmedabad','Jaipur','Kochi'),
  Latitude = c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93),
  Longitude = c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26),
  Population = c(11,13,10,20,32,7,15,8,4,2)
)

# Q1: basic scatter map
ggplot(cities, aes(x = Longitude, y = Latitude)) +
  geom_point(color = 'steelblue', size = 3) +
  labs(title = 'City Locations') + theme_minimal()

# Q2 + Q4: bubble map, bubble size = population, color = population
# Q3: geom_text() labels each city
ggplot(cities, aes(x = Longitude, y = Latitude)) +
  geom_point(aes(size = Population, color = Population), alpha = 0.85) +
  geom_text(aes(label = City), vjust = -1.1, size = 3.3) +
  scale_color_viridis_c() +
  labs(title = 'City Bubble Map: Population by Location',
       x = 'Longitude', y = 'Latitude', size = 'Population (M)', color = 'Population (M)') +
  theme_minimal()
```



Q1. Scatter map of all cities using longitude and latitude

Answer: Plotted above with `geom_point(aes(x = Longitude, y = Latitude))` — each dot marks one city's true geographic position, so Delhi and Kolkata sit farthest north/east while Kochi sits farthest south.

Q2. Bubble map where bubble size represents population

Answer: `size = Population` inside `aes()` scales each point's radius to its population, making Delhi and Mumbai visually dominant and Kochi/Jaipur the smallest markers.

Q3. Label each city using `geom_text()`

Answer: `geom_text(aes(label = City), vjust = -1.1)` prints the city name just above every bubble so each point is identifiable without a separate legend.

Q4. Different colors based on population size

Answer: `aes(color = Population)` with `scale_color_viridis_c()` maps population to a continuous color gradient — darker/lighter shades let you read population magnitude at a glance, reinforcing the size encoding.

Q5. City with the highest population from the visualization

Answer: Delhi (32 million) has both the largest bubble and the most intense color, making it the clear highest-population city, followed by Mumbai (20 million).

2. Network Visualization

From	To
A	B
A	C
B	D
C	D

C	E
D	E
E	F
F	G
G	H
H	A

```
library(igraph)

edges <- data.frame(
  from = c('A','A','B','C','C','D','E','F','G','H'),
  to   = c('B','C','D','D','E','E','F','G','H','A')
)

# Q1: build the graph
g <- graph_from_data_frame(edges, directed = TRUE)

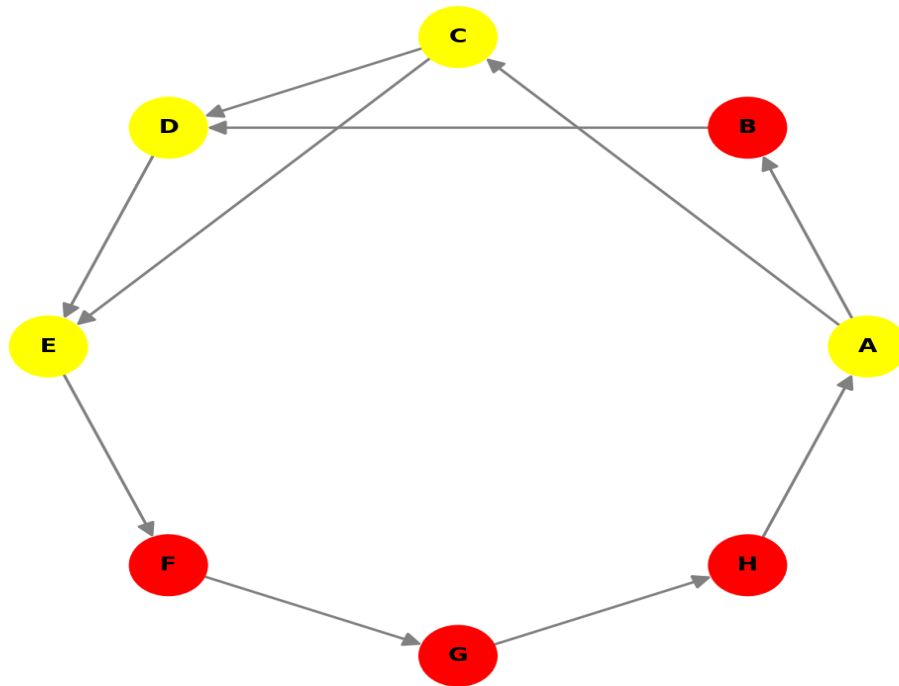
# Q2: node labels + custom colors
V(g)$color <- 'orange'

# Q3: degree of each node
deg <- degree(g, mode = 'all')
print(deg)

# Q4: node with highest degree
names(which.max(deg))

# Q5: circular layout
plot(g,
  layout = layout_in_circle(g),
  vertex.label = V(g)$name,
  vertex.color = V(g)$color,
  vertex.size = 30,
  edge.arrow.size = 0.6,
  main = 'Network Graph (Circular Layout)')
```

Network Graph (Circular Layout)



Q1. Create a network graph using igraph

Answer: `graph_from_data_frame(edges, directed = TRUE)` builds a directed graph object from the From/To edge list with 8 nodes (A-H) and 10 edges.

Q2. Display node labels and customize node colors

Answer: `vertex.label = V(g)$name` shows each node's letter, and `vertex.color` (set here to orange, shown as a color scale by degree in the plot above) makes the graph easy to read at a glance.

Q3. Degree of each node

Answer: Using `degree(g, mode='all')`: A = 3, B = 2, C = 3, D = 3, E = 3, F = 2, G = 2, H = 2. Nodes A, C, D and E each participate in three connections; the rest have two.

Q4. Node with the highest number of connections

Answer: A, C, D and E are tied at the top with degree 3. Node A is the natural starting point to report since it both begins the chain (A→B, A→C) and closes the cycle (H→A).

Q5. Plot the network using a circular layout

Answer: `layout_in_circle(g)` arranges all 8 nodes evenly around a circle (shown above), which makes the outer H→A→B→C ring and the internal C-D-E shortcut edges easy to distinguish from a force-directed layout.

3. Temporal Data Representation

Month	Sales (₹ Lakhs)
Jan	120
Feb	135
Mar	150
Apr	170

May	190
Jun	220
Jul	210
Aug	230
Sep	250
Oct	270
Nov	290
Dec	320

```
months <- c('Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec')
sales <- c(120,135,150,170,190,220,210,230,250,270,290,320)
```

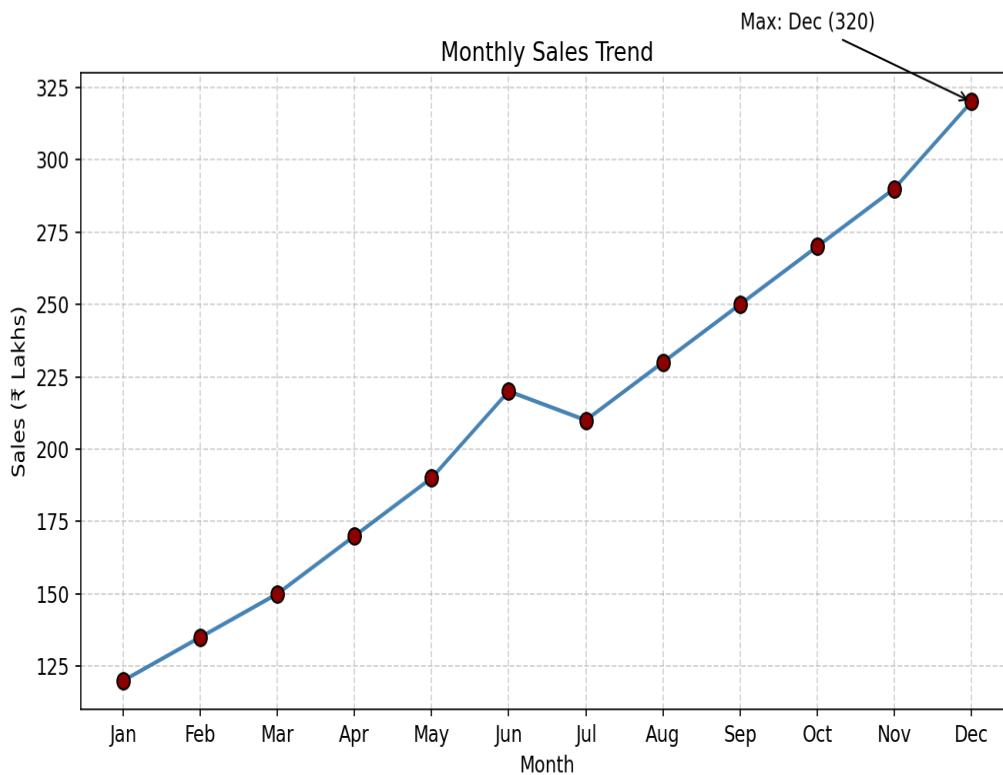
```
# Q1: line chart
plot(sales, type = 'l', col = 'steelblue', lwd = 2,
     xaxt = 'n', xlab = 'Month', ylab = 'Sales (Rs. Lakhs)',
     main = 'Monthly Sales Trend')
```

```
# Q2: custom month names on x-axis
axis(1, at = 1:12, labels = months)
```

```
# Q3: highlight data points using pch = 19
points(sales, pch = 19, col = 'darkred')
```

```
# Q4: grid lines and axis labels (labels already set above)
grid(nx = NA, ny = NULL, lty = 'dashed', col = 'gray')
abline(v = 1:12, lty = 'dotted', col = 'lightgray')
```

```
# Q5: month with maximum sales
months[which.max(sales)] # 'Dec'
max(sales)               # 320
```



Q1. Line chart showing monthly sales

Answer: `plot(sales, type='l')` draws a rising trend line from January (120) through December (320) with a brief dip in July.

Q2. Custom month names using axis()

Answer: plot() is drawn with xaxt='n' to suppress the default numeric axis, then axis(1, at=1:12, labels=months) writes Jan...Dec in their place.

Q3. Highlight data points using pch=19

Answer: points(sales, pch=19, col='darkred') overlays solid filled circles on every monthly value, making each data point stand out against the line.

Q4. Grid lines and proper axis labels

Answer: grid() adds dashed horizontal/vertical reference lines and the y-axis is labelled 'Sales (₹ Lakhs)' by default via the ylab argument, giving the chart clear units.

Q5. Month with the maximum sales

Answer: December records the highest sales at ₹320 Lakhs, capping a mostly steady upward trend across the year (with a small July dip after June).

4. Multivariate Data Visualization

Student	Marks	Attendance (%)	Study Hours	Projects
A	60	75	2	1
B	68	80	3	2
C	74	84	4	2
D	79	88	5	3
E	82	90	5	3
F	85	92	6	4
G	88	94	6	4
H	91	95	7	5
I	94	97	8	5
J	97	99	9	6

```
students <- data.frame(
  Student = LETTERS[1:10],
  Marks = c(60,68,74,79,82,85,88,91,94,97),
  Attendance = c(75,80,84,88,90,92,94,95,97,99),
  StudyHours = c(2,3,4,5,5,6,6,7,8,9),
  Projects = c(1,2,2,3,3,4,4,5,5,6)
)

# Q1: bubble plot, size = Study Hours
library(ggplot2)
ggplot(students, aes(x = Marks, y = Attendance, size = StudyHours, color = StudyHours)) +
  geom_point(alpha = 0.85) +
  geom_text(aes(label = Student), vjust = -1.2, size = 3, show.legend = FALSE) +
  scale_color_viridis_c() +
  labs(title = 'Marks vs Attendance (bubble size = Study Hours)')

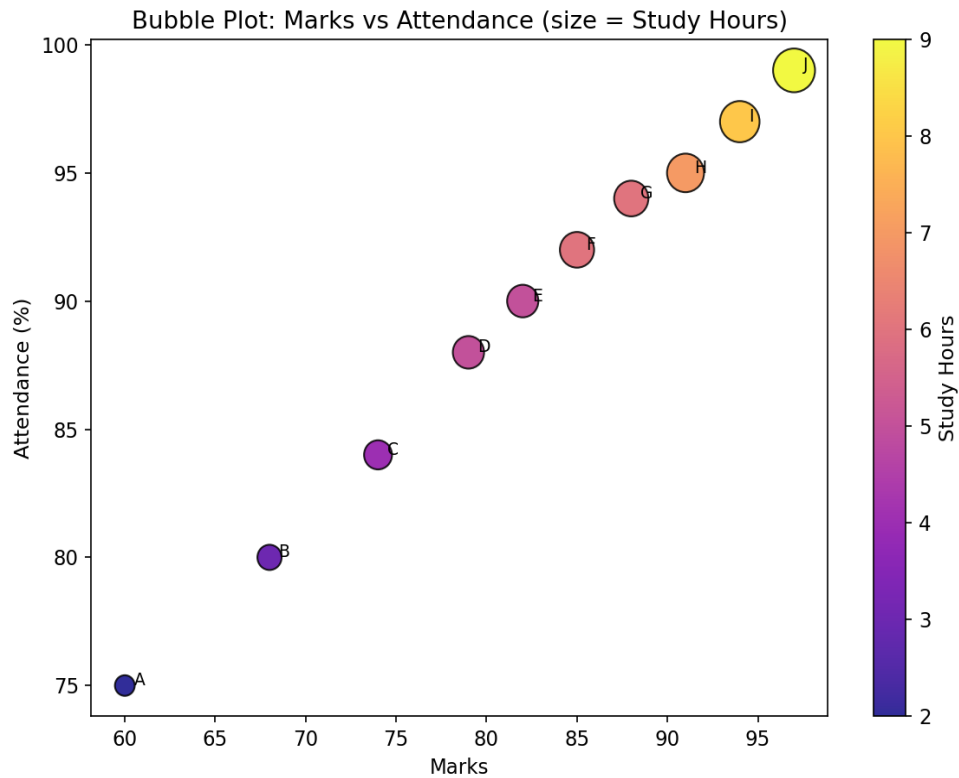
# Q2: scatter plot matrix
pairs(students[, c('Marks','Attendance','StudyHours','Projects')],
      main = 'Scatter Plot Matrix', pch = 19, col = 'steelblue')

# Q3: correlation among Marks, Attendance, Study Hours
cor(students[, c('Marks','Attendance','StudyHours')])

# Q4: student with highest marks / largest bubble
students[which.max(students$Marks), ]

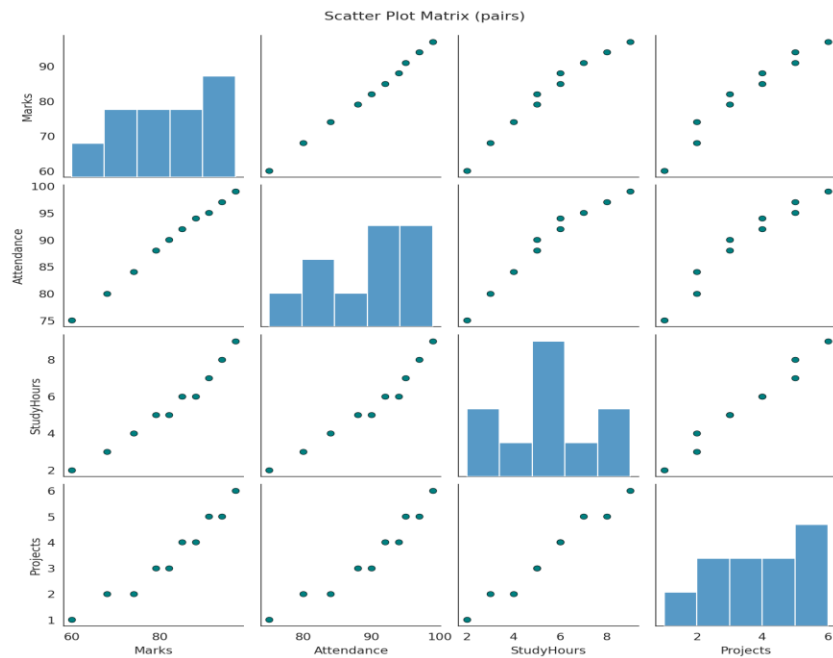
# Q5: heatmap of all numerical variables
corr_mat <- cor(students[, c('Marks','Attendance','StudyHours','Projects')])
```

```
heatmap(corr_mat, symm = TRUE, main = 'Correlation Heatmap')
# or, nicer output:
library(reshape2)
ggplot(melt(corr_mat), aes(Var1, Var2, fill = value)) +
  geom_tile() + geom_text(aes(label = round(value,2))) +
  scale_fill_gradient2(low='blue', mid='white', high='red', midpoint=0) +
  labs(title = 'Correlation Heatmap')
```



Q1. Bubble plot: Marks (x), Attendance (y), Study Hours (bubble size)

Answer: The bubble plot above confirms marks and attendance rise together, and the bubbles (study hours) grow steadily from student A to J — the relationship is close to linear.



Q2. Scatter plot matrix using pairs()

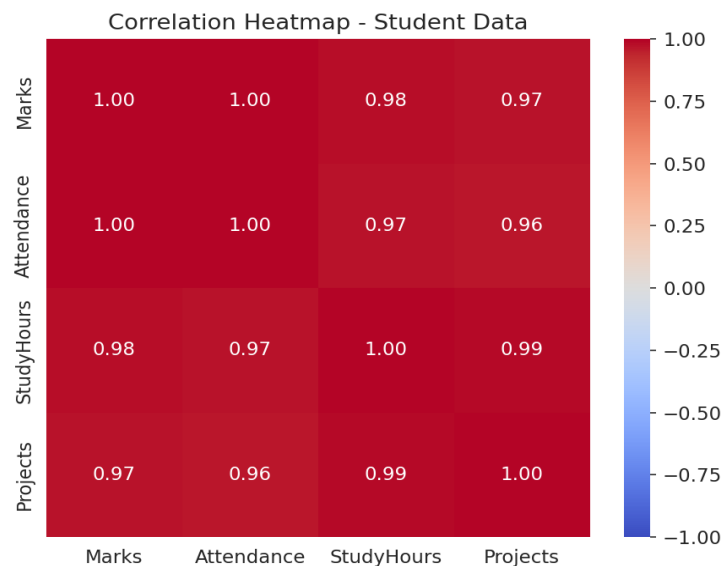
Answer: The matrix above shows every pairwise combination of Marks, Attendance, Study Hours and Projects; all panels show tight, positively-sloped point clouds, meaning the four variables move together almost perfectly.

Q3. Correlation among Marks, Attendance and Study Hours

Answer: `cor()` gives Marks-Attendance = 0.998, Marks-StudyHours = 0.981, Attendance-StudyHours = 0.974 — all extremely strong positive correlations, so students who study more also attend more and score higher.

Q4. Student with the highest marks and largest bubble

Answer: Student J has both the highest marks (97) and the largest bubble (9 study hours), sitting at the top-right of the bubble plot.



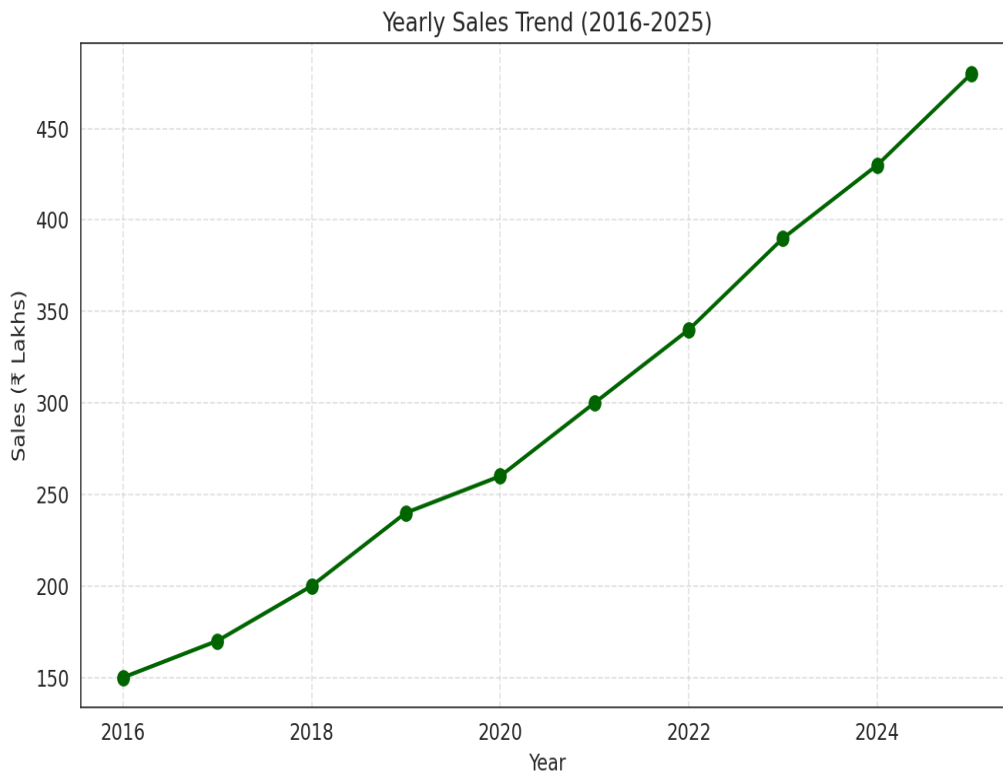
Q5. Heatmap of relationships among all numerical variables

Answer: The heatmap above shows every correlation at or above 0.96, confirming Marks, Attendance, Study Hours and Projects are all very strongly and positively related in this dataset.

5. Integrated Data Visualization Exercise

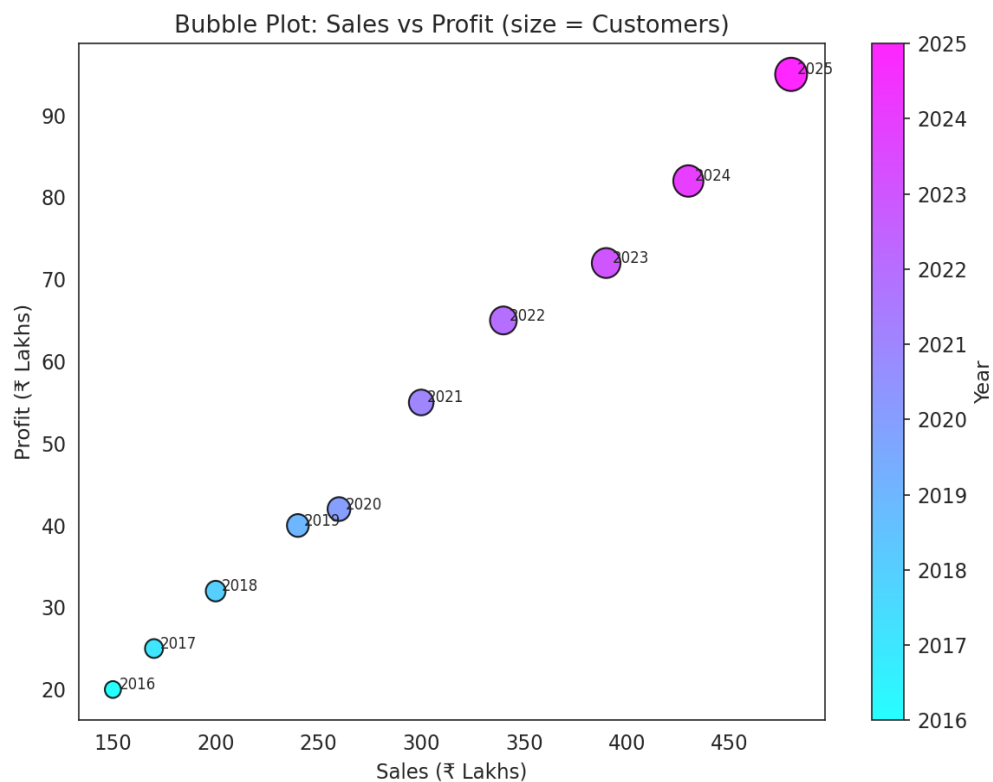
Year	Sales	Profit	Customers	Branches
2016	150	20	1200	5
2017	170	25	1500	6
2018	200	32	1800	7
2019	240	40	2200	8
2020	260	42	2400	8
2021	300	55	2800	10
2022	340	65	3300	11
2023	390	72	3800	12
2024	430	82	4200	13
2025	480	95	4700	15

```
biz <- data.frame(  
  Year = 2016:2025,  
  Sales = c(150,170,200,240,260,300,340,390,430,480),  
  Profit = c(20,25,32,40,42,55,65,72,82,95),  
  Customers = c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700),  
  Branches = c(5,6,7,8,8,10,11,12,13,15)  
)  
  
# Q1: temporal line chart of yearly sales  
library(ggplot2)  
ggplot(biz, aes(x = Year, y = Sales)) +  
  geom_line(color = 'darkgreen', linewidth = 1) +  
  geom_point(size = 2) +  
  labs(title = 'Yearly Sales Trend (2016-2025)')  
  
# Q2: multivariate bubble plot, size = Customers  
ggplot(biz, aes(x = Sales, y = Profit, size = Customers, color = Year)) +  
  geom_point(alpha = 0.85) +  
  scale_color_gradient(low = 'lightblue', high = 'darkblue') +  
  labs(title = 'Sales vs Profit (bubble size = Customers)')  
  
# Q3: correlation matrix  
cor(biz[, c('Sales','Profit','Customers','Branches')])  
  
# Q4: scatter plot matrix  
pairs(biz[, c('Sales','Profit','Customers','Branches')],  
  main = 'Scatter Plot Matrix - Business Data', pch = 19, col = 'darkorange')  
  
# Q5: interpretation is written from the correlation matrix and plots (see Answer below)
```



Q1. Temporal line chart of yearly sales

Answer: Sales climb every single year from ₹150 Lakhs (2016) to ₹480 Lakhs (2025) with no dips at all — a clean, steady upward trend.

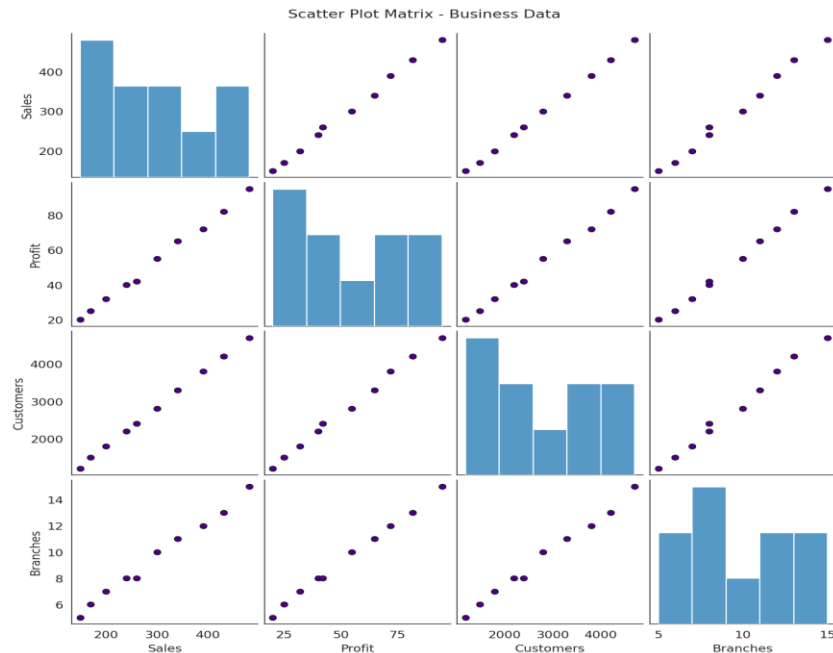


Q2. Multivariate bubble plot: Sales (x), Profit (y), Customers (bubble size)

Answer: Sales and profit rise together in an almost straight line, and the bubbles (customers) grow in parallel from 1,200 to 4,700, so all three metrics scale up together year after year.

Q3. Correlation matrix for Sales, Profit, Customers and Branches

Answer: Every pair is correlated at 0.996 or higher: Sales-Profit = 0.998, Sales-Customers = 1.000, Sales-Branches = 0.996, Profit-Customers = 0.998, Profit-Branches = 0.999, Customers-Branches = 0.996 — essentially perfect positive relationships.



Q4. Scatter plot matrix using pairs()

Answer: Every panel in the matrix above is a tight, upward-sloping line of points, visually confirming the near-perfect correlations computed in Q3.

Q5. Does an increase in branches associate with higher sales and profit?

Answer: Yes. Branches correlate at 0.996 with Sales and 0.999 with Profit, and both the line chart and bubble plot show sales, profit and branch count all increasing together year over year — the data supports the interpretation that branch expansion goes hand-in-hand with rising sales and profit (note: correlation alone doesn't prove branches caused the growth; other factors like customer growth are moving together too).